
Same-Evidence Reader-Budget Gating for Token-Efficient Agent Memory

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Abstract

Long-horizon agents can spend many reader tokens on redundant retrieved memories. We study a same-evidence reader-budget gate for the official open-source Mem0 package: the memory backend retrieves five records, and a deterministic risk rule chooses whether the frozen reader receives one or three of those exact records. On 311 LongMemEval questions in a low-accuracy constrained OSS setup, this adaptive gate answers 61/311 cases versus 62/311 for Mem0 top-5, while reducing reader tokens by 37.0% and reader-visible retrieved-context tokens by 53.3%. By comparison, fixed top-1 saves more tokens but falls to 50/311, fixed top-2 reaches 54/311, fixed top-3 reaches 60/311, and a matched-budget random top-1/top-3 route reaches 56/311. These results support a constrained token-savings claim for budgeted reader context, not a reproduction of Mem0 platform accuracy or a claim that the memory backend improves answer quality.

1. Introduction

Long-horizon agents increasingly depend on external memory to carry information across sessions, tools, and user interactions. Retrieval-centered memory systems are often evaluated by whether they recover useful evidence. A deployed frozen reader, however, also pays for every retrieved item it reads. Even when a memory backend retrieves relevant records, passing all of them to the reader can waste context on redundant, weakly relevant, or conflicting memories.

This paper asks a narrow token-budget question: under a fixed memory backend and frozen reader, how much of the retrieved evidence can be withheld before answer behavior changes? We keep the official open-source Mem0 package, reader model, prompt template, and decoding configuration fixed. The main intervention is not a new retriever. It is a same-evidence reader-budget gate that selects a smaller

prefix of the exact Mem0 top-5 records already retrieved for the query.

We evaluate on LongMemEval (Wu et al., 2024), a benchmark designed around long-term interactive memory, using the official Mem0 OSS package (Chhikara et al., 2025). The completed run covers knowledge-update, single-session-preference, multi-session, and single-session-user, for 311 questions. Official Mem0 top-5 spends 151,558 reader tokens for 62 correct answers. A deterministic adaptive gate routes low-risk cases to top-1 evidence and high-risk cases to top-3 evidence using only revision markers and answer-like values in the Mem0 records. It spends 95,466 reader tokens for 61 correct answers. This is the central result: an exact same-evidence budget controller removes 37.0% of reader tokens while changing the aggregate automatic correct count by one case in a constrained run where both policies fail most cases.

We also report ODV2-based secondary diagnostics because they motivated the budget-gating question. These are useful engineering checks, but the headline claim is deliberately based on the same-evidence adaptive gate because it avoids retrieval-overlap confounds. The result is not a benchmark-level Mem0 accuracy claim. It is a paired cost-reduction test under a constrained open-source Mem0 setup, and it preserves paired answer behavior rather than establishing high task quality: 244/311 cases are wrong for both top-5 and adaptive.

Contributions. We make four contributions. First, we define a same-evidence reader-budgeting interface that separates memory retrieval from the amount of retrieved evidence passed to a frozen reader. Second, we instantiate it for official Mem0 using a deterministic risk route over Mem0’s own top-5 records. Third, we provide paired LongMemEval evidence showing large reader-visible context reductions, while explicitly treating accuracy as a guardrail and reporting fixed top- k , matched-budget random routing, ODV2 compact, and ODV2-composed routes as secondary budget baselines. Fourth, we provide trace-level diagnostics of two reproducible failure modes—temporal-reasoning starvation and the stale-exposure paradox—that reveal a principled decomposition of long-horizon memory failures into state maintenance and evidence recall, and inform the conservative design of the risk route.

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2. Related Work

Retrieval-augmented generation shows that non-parametric memory can improve a frozen model by supplying external evidence (Lewis et al., 2020). Persistent-agent systems extend this idea to long-lived interaction. Generative Agents store observations and reflections (Park et al., 2023); MemGPT treats memory management as an operating-system-like problem (Packer et al., 2023); and Mem0 provides a scalable long-term memory substrate for production agents (Chhikara et al., 2025). These systems make external memory a practical adaptation mechanism, but they do not fully separate evidence recall from evidence budgeting.

Long-term memory benchmarks make that distinction important. LoCoMo evaluates very long-term conversational memory (Maharana et al., 2024), and LongMemEval focuses on interactive assistant memory across sessions (Wu et al., 2024). Many failures in these settings arise because evidence changes over time or because multiple retrieved memories are only partially relevant. Our work treats revision and conflicting answer values as routing signals for reader-budget control, rather than claiming to solve memory consolidation.

This design is related to belief revision (Alchourrón et al., 1985), but implemented as an engineering layer for LLM agents rather than a symbolic knowledge base. It is also aligned with efficient adaptation. Parameter-efficient tuning and continual-learning methods adapt model weights (Hu et al., 2022; Wang et al., 2024), whereas memory-state updates can be cheaper when the change is user-specific or short-lived. This is consistent with Green AI arguments for preferring lower-cost mechanisms when quality is preserved (Strubell et al., 2019; Schwartz et al., 2020).

3. Method

3.1. Same-Evidence Reader-Budget Gate

Let $B_5(q) = (r_1, \dots, r_5)$ be the five records returned by official Mem0 OSS for query q . The reader-budget gate never changes this retrieved set and never injects records from another store. It only chooses how many of the ranked records to expose to the frozen reader. Fixed baselines are therefore

$$B_k(q) = (r_1, \dots, r_k), \quad k \in \{1, 2, 3, 4, 5\}. \quad (1)$$

The adaptive gate uses a deterministic risk rule over Mem0’s own retrieved text. From the first four records, it counts revision markers such as *currently*, *recently*, *changed*, *updated*, and *used to*. It also extracts answer-like values with a numeric regex that covers money, times, percentages, durations, distances, and other digit-bearing values. The route is high-risk if at least one revision marker is present or at

least two distinct answer-like values are present. High-risk queries use $B_3(q)$; all other queries use $B_1(q)$:

$$B_{\text{ADAPT}}(q) = \begin{cases} B_3(q), & h(B_5(q)) = 1, \\ B_1(q), & h(B_5(q)) = 0. \end{cases} \quad (2)$$

This route uses no gold labels, predictions, category labels, or ODV2 records. It is a conservative reader-budget controller over a fixed retrieved evidence set.

```
REVISION = {now, currently, recently, new, changed,
            updated, revised, latest, moved, switched, instead,
            no longer, previously, formerly, used to, current}
ANSWER_VALUE_RE =
  r"(:?\$\\s*)?\\b\\d+(?:[.,:]\\d+)?"
  r"(:?\\s?(?:am|pm|kg|lb|lbs|%|percent|minutes?|\"
  r\"hours?|days?|weeks?|months?|years?|miles?|km))?\\b\"
high_risk = revision_count >= 1 or distinct_values >= 2
route = top3 if high_risk else top1
```

3.2. Secondary ODV2 Operating Points

We also report two ODV2 diagnostics because they motivated the budget-gating question. The compact ODV2 wrapper uses the same OSS Mem0 package and local reader, returns at most two records, and applies a conservative stale-state guard. The guard fires rarely in the completed run. Artifact inspection also shows that ODV2’s returned records are not always a subset of the corresponding official Mem0 top-5 records. We therefore treat ODV2 compact and the ODV2/top-3 composed route as secondary operating points, not as the causal basis for the same-evidence claim.

4. Evaluation

We evaluate on LongMemEval (Wu et al., 2024). The baseline is `official_mem0`, an adapter around the official open-source Mem0 package. The baseline retrieves five Mem0 records per query and passes all five to the frozen reader. We replay the same saved Mem0 records through the same reader using only the first $k \in \{1, 2, 3, 4\}$ records, then compose `official_mem0_same_evidence_adaptive` by routing each case to the already computed top-1 or top-3 replay. These same-evidence replays do not rerun Mem0 extraction, retrieval, or ODV2 state maintenance. We report paired comparisons on four LongMemEval categories: knowledge-update ($N = 78$), single-session-preference ($N = 30$), multi-session ($N = 133$), and single-session-user ($N = 70$), for 311 total questions.

The Mem0 extraction LLM is Qwen/Qwen2.5-7B-Instruct served through vLLM with a 16k serving context, bfloat16 weights, add batch size 16, and 2000 maximum message characters. Raw fallback is blocked as a valid outcome: if inferred Mem0 addition stores zero memories and

would fall back to raw storage, the run fails rather than producing an invalid result. The Mem0 embedder is `sentence-transformers/all-MiniLM-L6-v2` with local Qdrant storage. The frozen reader is also `Qwen/Qwen2.5-7B-Instruct`, evaluated with the repository local-Hugging Face backend, `bfloat16` inference, inference batch size 64, reader flush size 8, and an 8192-token reader context window. Reader decoding is deterministic greedy decoding with `do_sample=false`, `temperature=0`, `top_p=1.0`, `repetition_penalty=1.0`, `max_new_tokens=32`, and the tokenizer chat template enabled. Both policies use the same cleaned input file, `data/longmemeval_s_cleaned.json`, and the same reader prompt template. The run metadata records Python 3.12.3, PyTorch 2.5.1+cu121, and Transformers 4.47.1 on Linux 6.11 with CUDA 12.2. The artifact does not include a full installed package freeze; the repository lower bounds specify `mem0ai[nlp]>=0.1`, `qdrant-client>=1.9`, and `sentence-transformers>=3.0`.

Primary metrics are prompt tokens, reader total tokens, reader-visible retrieved-context tokens, and retrieved item count. Reader total tokens are Hugging Face tokenizer prompt tokens after chat-template rendering plus generated completion tokens. Reader-visible retrieved-context tokens are whitespace-token counts over the memory-record text passed to the reader; they measure reader context size, not backend retrieval work. Mem0 still retrieves the same top-5 records before the gate. Accuracy, paired wins, and paired losses against `official_mem0` are reported as guardrail metrics to show how much answer behavior changes under budget reduction. Pairing is by LongMemEval case id, so a win means the budgeted policy is correct where the baseline is incorrect, and a loss means the reverse. Accuracy uses a local span-match proxy rather than the official leaderboard protocol: normalized exact match or normalized span match after lowercasing, article removal, punctuation removal, and whitespace normalization. The results therefore should not be compared directly to Mem0 platform or leaderboard numbers, which use different Mem0 backends, larger retrieval depths, and different evaluation harnesses (?).

For secondary context, we also report `official_mem0.odv2_selective`, a compact ODV2 wrapper around the same OSS Mem0 setup, and `official_mem0.staleaware_gate`, a post-hoc composition of ODV2 compact and top-3 replay. These rows are not the basis of the same-evidence claim because ODV2 retrieval is not guaranteed to overlap exactly with the baseline top-5. As a secondary latency sanity check, we run a cache-free reader-only replay on a deterministic 64-case subset shared by official Mem0 top-5, the same-evidence adaptive row, and fixed top-1, top-2, and top-3 rows. This replay disables the response cache and reuses already-

retrieved records, so it checks that the token reduction is reflected in frozen-reader wall-clock time without rerunning Mem0 ingestion or extraction. Finally, we run an oracle answer-session replay on 64 cases by giving the reader only LongMemEval sessions marked as containing the answer.

Scope of the run. The experiment is intended as an efficiency ablation under a constrained OSS Mem0 reader budget, not as a reproduction of the published Mem0 platform result. The relevant question is paired: given the same OSS Mem0 extraction run and the same frozen reader, how much of the same retrieved context can be removed before accuracy changes? This design makes token savings directly interpretable, but it also explains the low absolute accuracy observed in the current run.

5. Results

Accuracy guardrail. Across the 311 paired cases, the automatic span scorer marks official Mem0 OSS top-5 correct on 62 cases and the same-evidence adaptive gate correct on 61 cases. The paired automatic difference is therefore one case, or -0.3 percentage points. There are 5 automatic paired wins for the adaptive gate and 6 automatic paired losses; 56 cases are correct for both methods and 244 are wrong for both. Among the 62 cases solved by top-5, adaptive preserves 56, or 90.3%, while using 37.0% fewer reader tokens. Because this one-case difference is sensitive to scorer boundary cases, we author-audited all 11 automatic disagreements and a deterministic 20-case agreement sample. This audit is author-confirmed, limited, and not blinded. The disagreement audit labels 5 cases correct for top-5 and 5 for adaptive, and the full 31-case worksheet labels both policies correct on 17 cases. The audit does not support an accuracy improvement claim, but it supports treating the automatic one-case difference as a guardrail rather than a substantive accuracy loss.

Reader-budget reduction. The token-savings signal is the central result. The adaptive gate reduces prompt tokens by 40.43%, total reader tokens by 37.01%, reader-visible retrieved-context tokens by 53.32%, and retrieved items passed to the reader by 53.35%. This is a clean same-evidence reader-budget result: Mem0 still retrieves the top-5 records, but every reader input is a prefix of the exact Mem0 top-5 records used by the baseline. The ODV2 compact row remains useful as a secondary operating point, reducing reader tokens by 42.70%, but it is not the main causal evidence because 63/311 ODV2 rows contain at least one record outside the corresponding official Mem0 top-5.

Cost-normalized utility. Because the goal is token savings under an accuracy guardrail, the most direct utility view is reader tokens spent per correct answer. Official Mem0 top-

Table 1. LongMemEval results for the exact same-evidence adaptive gate. Δ Prompt is adaptive gate minus official Mem0 top-5. Wins are cases where the adaptive gate is correct and Mem0 top-5 is wrong; losses are the reverse. Top2 is a fixed replay of the same Mem0 retrieved records with only the first two records passed to the reader.

CATEGORY	<i>N</i>	MEM0 ACC	ADAPTIVE ACC	TOP2 ACC	Δ PROMPT	WIN	LOSS
KNOWLEDGE-UPDATE	78	25/78 (.321)	25/78 (.321)	23/78 (.295)	−39.54%	1	1
SINGLE-SESS-PREF	30	0/30 (.000)	1/30 (.033)	0/30 (.000)	−43.69%	1	0
MULTI-SESSION	133	7/133 (.053)	7/133 (.053)	4/133 (.030)	−39.32%	2	2
SINGLE-SESS-USER	70	30/70 (.429)	28/70 (.400)	27/70 (.386)	−42.16%	1	3
TOTAL	311	62/311 (.199)	61/311 (.196)	54/311 (.174)	−40.43%	5	6

Table 2. Efficiency over the exact official Mem0 evidence frontier. Reader-token changes are relative to official Mem0 top-5. The adaptive row selects only between official Mem0 top-1 and top-3 replays; random 1/3 uses the same 105/206 top-1/top-3 route budget with seed 20260503.

POLICY	RECORDS	CORRECT	READER TOK.	Δ TOK.	TOK./CORRECT
TOP1	1	50/311	65,812	−56.58%	1,316
TOP2	2	54/311	86,002	−43.25%	1,593
RANDOM 1/3	1/3	56/311	95,230	−37.17%	1,701
ADAPTIVE	1/3	61/311	95,466	−37.01%	1,565
TOP3	3	60/311	110,058	−27.38%	1,834
TOP4	4	59/311	130,157	−14.12%	2,206
TOP5	5	62/311	151,558	—	2,444

Table 3. Mechanism and evidence-overlap audit. The headline adaptive gate uses only official Mem0 top-*k* replay rows; ODV2 rows are secondary because their returned records can differ from the baseline top-5.

DIAGNOSTIC	COUNT
SAME-EVIDENCE ADAPTIVE ROWS	311
ROUTES TO MEM0 TOP-1	105
ROUTES TO MEM0 TOP-3	206
MEM0 RECORDS OMITTED VS TOP-5	827
ODV2 ROWS WITH OUTSIDE-TOP-5 RECORD	63
ODV2 RECORDS OUTSIDE BASELINE TOP-5	111
STALE RECORDS REMOVED BY ODV2 GUARD	2

Table 4. Sensitivity of the same-evidence adaptive rule. The default route uses candidate-*k* = 4, revision threshold 1, and distinct-value threshold 2. All rows select only between official Mem0 top-1 and top-3 reader replays.

Setting	Route 1/3	Correct	Reader tok.	Δ
Default	105/206	61/311	95,466	−37.01%
Less	137/174	60/311	91,039	−39.93%
More	87/224	61/311	97,962	−35.36%
Value	126/185	60/311	92,393	−39.04%
Broad	70/241	61/311	100,415	−33.74%

5 spends 151,558 reader tokens for 62 correct answers, or 2,444 reader tokens per correct answer. The same-evidence adaptive gate spends 95,466 reader tokens for 61 correct answers, or 1,565 tokens per correct answer. Equivalently, the top-5 baseline spends 56,092 additional reader tokens to recover one additional aggregate correct answer. Fixed top-1 has the lowest token cost per correct answer but answers only 50 cases correctly; fixed top-2 saves slightly more tokens than the adaptive gate but answers seven fewer cases correctly.

A paired bootstrap over 20,000 case resamples gives a 95% interval of 35.5–38.5% for the adaptive reader-token reduction and 28.5–42.7% for tokens-per-correct reduction. The same bootstrap gives an accuracy-delta interval of −2.57 to +1.61 percentage points. We do not claim statistical non-inferiority for any budgeted row. The same-evidence adaptive result should be read as an accuracy-guarded oper-

ating point, not as an accuracy improvement.

Scorer sanity checks. The adaptive guardrail audit is the scorer check for the headline comparison. We also retain the earlier 50-case audit from the original ODV2 comparison only as secondary scorer context, not as evidence for the headline adaptive policy. That audit agrees with the automatic span scorer on 91/100 decisions across official Mem0 and ODV2 compact. Together, these checks support using the local span scorer as a rough guardrail, while preserving the paper’s narrower claim: token savings under observed paired answer behavior, not statistically certified non-inferiority.

All sensitivity rows are evaluated on the same completed run and should be read as robustness checks around the deterministic rule, not as evidence of pre-registered model selection.

Top- k attribution baselines. Tables 2 and 3 show that fixed truncation alone is not the whole story. Plain top-1 replay saves the most tokens but drops to 50/311 correct. Plain top-2 uses 86,002 total reader tokens but answers only 54/311 cases correctly. Plain top-3 recovers 60/311 correct answers, but pays 110,058 reader tokens. To test whether adaptive’s benefit is only the top-1/top-3 budget mix, we also sample matched-budget random routers with the same 105/206 top-1/top-3 route counts. The deterministic random route with seed 20260503 answers 56/311 at 95,230 reader tokens. Across 10,000 random matched routes, the correct-count median is 57 and the 95% interval is 53–60; adaptive’s 61 correct answers are greater than or equal to 99.57% of random matched draws. This percentile is exploratory and not a pre-registered hypothesis test. It suggests the risk route adds signal beyond budget mixing, although it remains same-run evidence rather than prospective validation.

Absolute accuracy caveat. The absolute accuracy values are low, especially in single-session-preference and multi-session. This is not hidden in the result tables because it directly affects how the paper should be read. The experiment is a paired efficiency ablation under one constrained OSS Mem0 setup and one local reader, not a claim that either method reaches published Mem0 platform accuracy. The oracle answer-session sanity check reaches 27/64 correct (42.2%) on its 64-case sample, including 10/12 single-session-user cases, showing that the same reader and scorer can solve substantially more cases when focused answer evidence is supplied. The oracle result remains far from saturated, so it should be read only as a sanity check on evidence availability and scorer strictness.

Uncertainty and latency sanity check. The paired accuracy differences are small relative to the number of cases, and we do not claim statistical non-inferiority. We keep reader-token count as the primary efficiency metric because it is deterministic from the generated prompts and directly controls hosted-reader cost. In the cache-free 64-case reader replay, official Mem0 top-5 takes 62.6 ms/case, while the same-evidence adaptive row takes 37.2 ms/case. Fixed top-1, top-2, and top-3 take 26.2, 33.4, and 39.6 ms/case, respectively. Adaptive answers 10/64 cases correctly on this subset versus 9/64 for top-5, top-2, and top-3; top-1 also answers 10/64. Peak allocated GPU memory is 16,179 MiB for top-5 and 15,686 MiB for adaptive. This is a reader-only latency sanity check over saved records, not a deployment benchmark; the present run still does not report energy or end-to-end wall-clock cost across heterogeneous hardware.

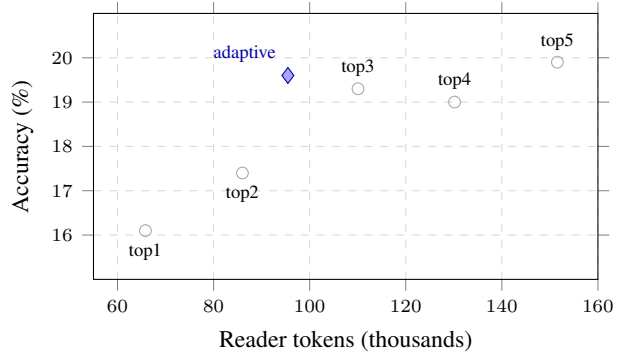


Figure 1. Accuracy–cost tradeoff for exact official Mem0 evidence only. The blue marker is the same-evidence adaptive gate; gray markers are fixed Mem0 top- k replays.

6. Trace-Level Diagnostics of Budget-Gating Failures

The risk-routing signal in the adaptive gate was motivated in part by secondary ODV2 diagnostics that surface two reproducible failure modes when retrieved evidence is either compacted or filtered without preserving revision context. We present anonymised reproducible triggers for each mode. Because ODV2 retrieval is not guaranteed to overlap exactly with the baseline top-5 records (Table 3), these diagnostics inform gate design rather than constitute same-evidence results.

Temporal-reasoning starvation. Consider a multi-turn conversation in which a user mentions restaurants across several sessions: first recommending Sakura, then switching to Noma, briefly returning to Sakura, then settling on Noma again. The question is: “Which restaurant did the user mention first?” Mem0 retrieves all four mentions in chronological order, giving the reader the full trace needed to answer “Sakura.” A compaction pass that deduplicates the two Sakura mentions and the two Noma mentions into one entry each reduces the four-entry trace to two. The compacted context no longer preserves temporal ordering; the reader answers “Noma”—the currently active preference—rather than “Sakura.” The compaction was semantically correct for a current-state query but destructive for a history query.

This failure mode directly motivates the adaptive gate’s conservative top-3 route for high-risk cases: when revision markers are present, the gate retains more context rather than aggressively truncating, preserving temporal ordering at modest extra cost.

Stale-exposure paradox. Consider a conversation in which a user states they enjoy hiking (turn 3), then later switches to cycling (turn 47). The question is: “Does the user still enjoy hiking?” Mem0 retrieves both the hiking

entry (archived, superseded) and the cycling entry (active). The frozen reader sees both, recognises the revision, and correctly answers “No, they switched to cycling.” A filter that silently removes the superseded hiking entry—because the current cycling entry is already present—leaves the reader with only “the user enjoys cycling.” The reader answers “No” for the wrong reason: it has no evidence the user ever enjoyed hiking and cannot confirm or deny the implicit history. The expected answer includes the contrast (“they used to, but switched to cycling”), so this scores as a loss.

The fix is revision-marker injection rather than silent removal: instead of dropping the superseded entry, the policy injects a compact marker such as [previously: hiking, now: cycling] that preserves the contrast without surfacing the full stale record. This informs a key design choice of the adaptive gate: the gate does *not* attempt to filter stale records from the Mem0 top- k prefix. It only controls how many ranked records reach the reader, leaving staleness resolution to the reader when revision signals are present.

7. Discussion

The current results support a narrower decomposition of agent memory into evidence recall and reader-budget control. Official Mem0 OSS supplies the retrieved evidence, while the same-evidence gate decides how much of that ranked evidence is passed to the frozen reader. The adaptive route substantially reduces the amount of text read by the answerer while staying within one aggregate correct answer of the top-5 baseline. This distinction should remain central to the paper framing.

This framing matters for cost-aware agent memory. If every memory query requires passing many retrieved facts to a reader, long-horizon agents inherit a growing prompt-cost burden even when the backend can retrieve relevant evidence. The reported run shows that a lightweight post-retrieval gate can remove over half of reader-visible retrieved-context tokens under a fixed open-source Mem0 setup. The adaptive gate is best used as a budgeted utility ablation: it reduces reader tokens per correct answer from 2,444 to 1,565, improves substantially over fixed top-1 and top-2 in accuracy, and uses fewer tokens than fixed top-3 while answering one more case correctly.

State maintenance versus evidence recall. The trace-level diagnostics in Section 6 reveal a deeper decomposition of long-horizon agent memory failures into two distinct jobs that prior memory systems often bundle together. The first is *state maintenance*: deciding which value is currently active after revisions, reversions, and scope splits. The second is *evidence recall*: recovering the right conversational span when the task expects supporting context rather than only the

final value. Stale-state exposure is fundamentally a failure of state maintenance, while temporal-reasoning starvation is a failure of evidence recall. The adaptive budget gate addresses neither problem directly—it only controls how many ranked records reach the reader—but the risk signal is explicitly calibrated to avoid triggering on queries that require the full revision trace, making the gate safer on evidence-recall tasks than aggressive fixed truncation.

Deployment implications. The cost-accuracy tradeoff reported here has practical boundaries. If an application is dominated by stateful facts—user profile changes, current preferences, or evolving tool state—a validity-aware filter like ODV2-selective is attractive because it collapses redundant revision history while preserving current state. If the application frequently requires citation-like support from prior dialogue, episodic traces, or event descriptions, the Mem0 baseline is safer because realistic conversational QA rewards high-recall evidence. The adaptive budget gate occupies a middle position: it preserves more evidence than fixed top-1 or top-2 on high-risk queries while still reducing reader cost substantially on low-risk queries. This makes it most useful in mixed-domain deployments where staleness and budgeting pressure co-occur.

Relation to multimodal agent memory. The budget-gating design is relevant beyond text-only settings. A multimodal agent may receive updates about the same entity across modalities—location via text, preferences via image interaction, schedule via calendar events. The revision signals used by the adaptive gate (update markers, distinct answer-like values) operate on the text of retrieved records and are indifferent to the modality from which those records were originally extracted. A multimodal memory system that produces structured records with similar fields could use the same risk route without modification. The state-maintenance / evidence-recall decomposition also provides a clean evaluation framework for multimodal memory: tasks requiring current-state correctness and tasks requiring episodic trace recovery call for architecturally different solutions and should be benchmarked separately.

The low absolute accuracy is also informative. It indicates that this run is not a reproduction of the published Mem0 platform benchmark configuration. In our setup, official Mem0 OSS retrieves five records and the reader is a local Qwen-7B model scored by a strict local span-match proxy. Published Mem0 platform results use different infrastructure and much larger retrieval depths. The manual audit reinforces this interpretation: several wrong answers are better described as missing retrieved evidence than reader arithmetic failures. The proper comparison inside this paper is therefore paired and local: official Mem0 OSS top-5 versus smaller prefixes of the same retrieved Mem0 evidence.

8. Limitations

The main limitation is that the result does not show an accuracy improvement. Accuracy is not the target metric of the paper; it is a guardrail for the token-savings claim. We therefore do not claim that the adaptive gate beats Mem0, official Mem0, or the Mem0 platform on broad conversational QA. The defensible result is narrower: same-evidence adaptive gating is a lower-cost operating point that reduces reader tokens by 37.0% and reader tokens per correct answer by 36.0% relative to the top-5 official Mem0 baseline, while losing one aggregate correct answer.

Second, the baseline is the official open-source Mem0 package under a constrained retrieval budget, not the Mem0 managed platform and not the published top-50 or top-200 benchmark setting. The low absolute accuracy should be reported rather than hidden, but it should not be interpreted as a claim about Mem0 platform quality. The top- k replay study covers $k \leq 5$ from the completed run, but a stronger version should compare top-10, top-50, and top-200 retrieval to separate retrieval recall failures from reader failures.

Third, the adaptive route is deterministic but not pre-registered; it should be validated prospectively in a fresh run. The matched-budget random control reduces the concern that the result is only a budget-mixing artifact, but it is still derived from the same completed run. The adaptive route also composes existing top-1 and top-3 replay outputs rather than measuring a single online controller with its own retrieval path. The cache-free row measures reader latency for saved adaptive inputs, but not Mem0 retrieval time, routing overhead, or serving under concurrent traffic. Fourth, the adaptive guardrail audit is author-confirmed and targeted, not blinded or independently replicated. It checks the fragile disagreement cases but does not replace a full human evaluation of all 311 answers. Fifth, token count is a proxy for reader cost, not a complete deployment metric. This is appropriate for the paper’s token-savings claim, but future deployment work should add repeated latency trials, GPU utilization, energy, and wall-clock cost. A sixth limitation concerns the two failure modes surfaced by the ODV2 secondary diagnostics. Temporal-reasoning queries require the full chronological trace; any compaction or truncation that destroys temporal ordering degrades accuracy on these cases regardless of the budget level. The adaptive gate mitigates this by routing high-risk cases to top-3 rather than top-1, but it does not fully resolve it: a dedicated query-mode classifier that routes history queries to the untruncated top-5 would be necessary for robust temporal reasoning. Similarly, the stale-exposure paradox shows that silently withholding superseded evidence can hurt on comparative questions; revision-marker injection is a cleaner fix than budget-level control alone, but implementing it requires changes to the prompt template that are outside the

same-evidence constraint of the current design.

Finally, the current experiments are text-only. The SCALE setting motivates multimodal agent memory, but this paper only tests the text memory layer that a multimodal extraction system could feed.

9. Conclusion

Same-evidence adaptive gating provides evidence that agent memory systems can reduce frozen-reader token budgets without changing the memory backend. On 311 paired LongMemEval questions, the adaptive gate reduces reader total tokens by 37.0% and reader-visible retrieved-context tokens by 53.3%, while the automatic accuracy guardrail changes from 19.9% to 19.6% and the author-confirmed disagreement audit is tied. Fixed top-2 reaches a lower token budget but lower accuracy, 17.4%, while fixed top-3 reaches 19.3% at a larger budget and matched-budget random routing reaches 18.0%. The main contribution is not a universal retriever or a new benchmark-leading memory system. It is a clean, same-evidence budget ablation showing that a frozen reader can often receive fewer retrieved memories while preserving most paired answer behavior in this constrained Mem0 OSS setup.

LLM/Agent Usage Disclosure

LLM-based coding and writing tools were used to assist with implementation and reviewer-style critique. The authors remained responsible for experiment execution, numerical results, claim selection, and final manuscript content.

Artifact Status

The submitted repository includes the official Mem0 package runner, top- k replay, same-evidence adaptive composition, matched-budget random routing, token analysis, submission-check generation, cache-free reader benchmarking, oracle answer-context replay, and per-case JSONL diagnostics. The active run directory is `results/official_mem0_basecompact_full_20260502T072459Z`. It contains the top- k replay rows, adaptive rows, random matched-control rows, cleaned-input checksum, command logs, vLLM/server diagnostics, and GPU snapshots. Submission-check outputs include the adaptive guardrail audit, reviewed 50-case scorer audit, oracle answer-context rows, and cache-free reader traces.

Reproducibility Commands

The adaptive rows are regenerated with `scripts/compose_official_mem0_same_evidence_`

`adaptive.py`; the route uses candidate- $k = 4$, revision-signal threshold 1, and distinct-value threshold 2. The matched-budget random control is regenerated with `scripts/compose_official_mem0_random_matched_route.py`. The paper tables are regenerated from per-case JSONL outputs using the repository scripts named above. The completed run stores command logs, diagnostics, top- k replays, bootstrap summaries, adaptive guardrail audit, random matched-control distribution, manual-audit sample, cache-free reader benchmark, and oracle answer-context replay under the active result directory.

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